characters in its label to the next characters in the string. Moreover, binary search allows us to find this edge in logarithmic time in the number of edges, and since there can't be more than $k=|\sum|$ edges per node (because there can be at most one starting with each character in our alphabet $\sum$), we know that the worst case running time for performing binary search and finding a candidate edge is $O(\log(k))$.

Since k is a constant that doesn't depend neither on the number of keys stored in the trie nor on the length of the words searched/inserted/etc..., we can consider $O(\log(k))=O(1)$ as far as asymptotic analysis is concerned. Moreover, no extra space[18] is required to store edges, with this solution.

